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Abstract

Online controlled experiments are at the heart of making data-driven decisions at a diverse set of companies, including Amazon, eBay, Facebook, Google, Microsoft, Yahoo, and Zynga. Small differences in key metrics, on the order of fractions of a percent, may have very significant business implications. At Bing it is not uncommon to see experiments that impact annual revenue by millions of dollars, even tens of millions of dollars, either positively or negatively. With thousands of experiments being run annually, improving the sensitivity of experiments allows for more precise assessment of value, or equivalently running the experiments on smaller populations (supporting more

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Improving the Sensitivity of Online Controlled Experiments by Utilizing Pre-Experiment Data

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ABSTRACT

Online controlled experiments are at the heart of making data-driven decisions at a diverse set of companies, including Amazon, eBay, Facebook, Google, Microsoft, Yahoo, and Zynga. Small differences in key metrics, on the order of fractions of a percent, may have very significant business implications. At Bing it is not uncommon to see experiments that impact annual revenue by millions of dollars, even tens of millions of dollars, either positively or negatively. With thousands of experiments being run annually, improving the sensitivity of experiments allows for more precise assessment of value, or equivalently running the experiments on smaller populations (supporting more experiments) or for shorter durations (improving the feedback cycle and agility). We propose an approach (CUPED) that utilizes data from the pre-experiment period to reduce metric variability and hence achieve better sensitivity. This technique is applicable to a wide variety of key business metrics, and it is practical and easy to implement. The results on Bing’s experimentation system are very successful: we can reduce variance by about 50%, effectively achieving the same statistical power with only half of the users, or half the duration.

Categories and Subject Descriptors

Keywords

Controlled experiment, Variance, A/B testing, Search quality evaluation, Pre-Experiment, Power, Sensitivity

1. INTRODUCTION

A controlled experiment is probably the oldest and the most widely accepted methodology to establish a causal relationship (Mason et al. 1989; Box et al. 2005; Keppel et al. 1992; Kohavi et al. 2009b; Manzi 2012). Now widely used and re-discovered by many companies, it is referred to as a randomized experiment, an A/B test (Wikipedia), a split test, a weblab (at Amazon), a live traffic experiment (at Google), a flight (at Microsoft), and a bucket test (at Yahoo!). This paper is focused on online controlled experiments, where experiments are conducted on live traffic to support data-driven decisions in online businesses, including e-business sites like Amazon and eBay, portal sites like Yahoo and MSN (Kohavi et al. 2009b), search engines like Microsoft Bing (Kohavi et al. 2012) and Google (Tang et al. 2010b).

Online controlled experiments are critical for businesses. Small differences in key metrics, on the order of fractions of a percent, can have very significant business implications. At Bing it is not uncommon to see experiments that im-